Mina Gabriel

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Summary

Data Scientist and AI/ML engineer with experience leading applied research, evaluating intelligent systems, and developing data-driven software solutions across public-sector, academic, and enterprise environments. Background includes artificial intelligence, machine learning, NLP, retrieval-augmented generation (RAG), vector databases, data analysis, model evaluation, data visualization, research design, and scalable software platforms. Experienced in developing data management practices, improving data quality and auditability, evaluating system performance, and translating analytical findings into technical recommendations for stakeholders and leadership. Proven record delivering AI, data science, and software engineering solutions in government, research, and industry settings, including the Commonwealth of Pennsylvania, Cisco, NASA/JPL, Harrisburg University, and Temple University.

Core Skills

- **Programming:** Python, Java, Kotlin, C#, VB.NET, JavaScript/TypeScript, Node.js
- **Data Platforms:** PostgreSQL, SQL Server, MySQL, Redis, MongoDB, Snowflake, Spark, vector databases; schema design, normalization, query optimization, data quality, performance tuning
- **Machine Learning & AI:** PyTorch, TensorFlow, OpenCV, model training and serving, data pipelines, explainable AI, Generative AI/NLP, computer vision, model evaluation
- **Research & Analytics:** experimental design, statistical/empirical evaluation, uncertainty estimation, RAG evaluation, simulation workflows, technical reporting, stakeholder briefings
- **Cloud & DevOps:** AWS, Azure, Docker, Kubernetes, GitHub Actions, Jenkins, logging, metrics, tracing
- **Backend & Distributed Systems:** RESTful and SOAP APIs, microservices, GraphQL, OAuth 2.0, JWT, message-driven and event-streaming systems
- **Security:** encryption in transit and at rest, identity management, secure communication protocols
- **Languages:** English (fluent), Arabic (fluent)

Experience

- **Commonwealth of Pennsylvania** — Harrisburg, PA
  Senior Applications Developer — *Jan 2026 – Present* — *37.5 hrs/week*
  - Lead technical delivery for CWDS / PA CareerLink, an enterprise workforce platform supporting multiple Commonwealth agencies and program bureaus, with responsibility for system reliability, data integrity, auditability, and production readiness.
  - Lead development of enterprise data management practices supporting operational goals, including data quality review, traceability, reporting accuracy, SQL Server optimization, and production support for high-visibility public-sector workflows.
  - Lead the full software development lifecycle for complex initiatives, including requirements analysis, solution evaluation, technical specification development, implementation, release coordination, and post-deployment support.
  - Develop and maintain .NET/C# applications using ASP.NET MVC and Web APIs, integrating internal and external services to support secure, scalable, and dependable program operations.
  - Communicate technical risks, system behavior, implementation options, and production findings to developers, analysts, business stakeholders, vendor teams, and management to support reliable operational decisions.
- **Harrisburg University of Science and Technology** — Harrisburg, PA
  Lecturer, Computer Science — *Aug 2014 – Jan 2026* — *40 hrs/week*
  - Lead AI, data science, and software engineering instruction across undergraduate and graduate courses in Machine Learning, Computer Vision, Advanced Databases, Algorithms, Data Structures, and Mobile Computing.
  - Lead applied research and experiential learning projects involving data pipelines, databases, APIs, machine learning models, computer vision, observability, and production-oriented software systems.
  - Design technical curriculum, guide capstone teams and undergraduate research, evaluate analytical results, and present complex computing concepts to students, faculty, industry partners, and academic leadership.

- **Cisco** Remote
  Software Engineer III (Research & Engineering) *Oct 2022 – May 2023* — *20 hrs/week*
  - Led applied AI research and engineering workstreams for an intelligent perception system integrating multi-object tracking, speech-to-text, and on-device inference with TensorFlow Lite and OpenCV.
  - Designed and implemented a hybrid reasoning layer combining rule-based logic with large language model integration to reduce false positives and prioritize events under real-time, compute-constrained conditions.
  - Deployed containerized microservices to AWS (ECR/ECS, EC2, S3), enabling scalable experimentation and evaluation of model performance, latency, reliability, and cost tradeoffs through CI/CD pipelines.
  - Engineered high-throughput, low-latency data pipelines using Redis for feature caching and real-time processing, supporting repeatable experimentation, operational data flow, and downstream analytical use.
- **NASA (JPL & Caltech collaboration)** Remote
  Research Assistant & Software Engineer (Contract) *Oct 2020 – Sep 2021* — *20 hrs/week*
  - Led applied AI R&D tasks for a hybrid perception–reasoning architecture combining deep neural networks with the Non-Axiomatic Reasoning System (NARS) for explainable decision-making under uncertainty.
  - Designed experimental workflows for object detection and tracking using YOLOv4, SORT/Kalman-filter methods, simulated data, and custom datasets to evaluate accuracy, speed, and robustness.
  - Architected modular perception, tracking, reasoning, and user-interaction components to reduce end-to-end latency, support scalable experimentation, and improve real-time system performance.
  - Built and deployed a cross-platform mobile application with an AI chatbot, hazard sign recognition, voice interaction, and real-time mapping to deliver on-device situational awareness.
  - Deployed containerized components on AWS to support telemetry collection, performance evaluation, continuous learning, and post-deployment analysis across simulated safety scenarios.
- **Temple University** Philadelphia, PA
  Research Assistant, Artificial Intelligence *Jan 2021 – Dec 2021* — *40 hrs/week*
  - Conducted AI research with Dr. Pei Wang on the Non-Axiomatic Reasoning System (NARS), focusing on adaptive logic, uncertainty, temporal reasoning, and decision-making under limited resources.
  - Developed and evaluated OpenNARS-inspired components in Node.js/TypeScript, including task scheduling, evidence accumulation, temporal inference, and reasoning processes for intelligent systems.
  - Investigated integration of reasoning with perception and language modules, contributing to research on explainable, real-time, and resource-bounded artificial intelligence.
- **The APAK Group - Hershey Center for Applied Research** Hummelstown, PA
  Software Engineer (Full-Stack) *Jan 2012 – Feb 2015* — *40 hrs/week*
  - Developed full-stack industrial automation applications using .NET/C# (ASP.NET), JavaScript, and SQL, including normalized database schemas, stored procedures, and real-time dashboards.
  - Designed system architectures and HMIs integrating distributed control systems, peripheral hardware, and industrial network protocols to improve operator efficiency and system usability.
  - Optimized and validated system performance through debugging, on-site commissioning, FactorySQL integration, role-based access, auditing, error handling, and data integrity controls.
- **I2I Vision** Alexandria, Egypt
  Web Developer & Database Administrator *Jan 2009 – Aug 2010* — *40 hrs/week*
  - Developed LAMP-stack applications using PHP, MySQL, JavaScript, HTML, and CSS; managed database backups, indexing, query performance, and data reliability.
  - Implemented REST endpoints and administrative dashboards; improved data integrity through constraints, triggers, migration scripts, and operational database support.

Education

- **Temple University** Philadelphia, PA
  Ph.D. in Computer & Information Sciences (Artificial Intelligence), in progress *Jan 2020 – Present*
- **Harrisburg University of Science and Technology** Harrisburg, PA
  M.S., Information Systems Engineering & Management *Jan 2012 – Apr 2014*
- **Arab Academy for Science, Technology & Maritime Transport (AAST)** Alexandria, Egypt
  B.S., Computer Science *Jan 2004 – Aug 2008*